

Chat Link: <https://share.gemini.google/TumguaUVwXKU>

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{
  "global_data": {
    "attribute_1": "total_cost",
    "attribute_2": "material_continuity",
    "attribute_3": "power_density",
    "attribute_4": "thermal_performance",
    "attribute_5": "manufacturability",
    "direction_attribute_1": "negative",
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    "direction_attribute_3": "positive",
    "direction_attribute_4": "positive",
    "direction_attribute_5": "positive",
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    "attribute_2_max": 1.00,
    "attribute_3_min": 0.45,
    "attribute_3_max": 0.80,
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    "attribute_4_max": 210.0,
    "attribute_5_min": 0.60,
    "attribute_5_max": 0.98
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      "copper_price_per_kg": 9.80,
      "aluminum_price_per_kg": 2.15,
      "copper_product_market_share": 0.98,
      "aluminum_product_market_share": 0.02,
      "weight_attribute_1": 95,
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      "weight_attribute_3": 85,
      "weight_attribute_4": 75,
      "weight_attribute_5": 90
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    {
      "region": "United States",
      "copper_price_per_kg": 9.50,
      "aluminum_price_per_kg": 2.80,
      "copper_product_market_share": 1.00,
      "aluminum_product_market_share": 0.00,
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    "weight_attribute_1": 75,
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    "region": "Europe",
    "copper_price_per_kg": 9.55,
    "aluminum_price_per_kg": 2.65,
    "copper_product_market_share": 0.99,
    "aluminum_product_market_share": 0.01,
    "weight_attribute_1": 70,
    "weight_attribute_2": 75,
    "weight_attribute_3": 90,
    "weight_attribute_4": 95,
    "weight_attribute_5": 65
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  {
    "region": "India",
    "copper_price_per_kg": 10.10,
    "aluminum_price_per_kg": 2.55,
    "copper_product_market_share": 0.92,
    "aluminum_product_market_share": 0.08,
    "weight_attribute_1": 100,
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    "weight_attribute_3": 50,
    "weight_attribute_4": 55,
    "weight_attribute_5": 60
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"product_data": [
  {
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    "dominant material": "cu",
    "product": "EV Traction Motor Winding",
    "copper_kg": 8.5,
    "aluminum_kg": 0.0,
    "total_mass_kg": 9.2,
    "non_material_cost_per_unit": 35.00,
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    "attribute_4_value": 155.0,
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  "region": "China",
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  "total_mass_kg": 4.8,
  "non_material_cost_per_unit": 38.00,
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  "attribute_3_value": 0.52,
  "attribute_4_value": 135.0,
  "attribute_5_value": 0.74
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  "product": "EV Traction Motor Winding",
  "copper_kg": 12.0,
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  "total_mass_kg": 13.1,
  "non_material_cost_per_unit": 65.00,
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  "attribute_3_value": 0.75,
  "attribute_4_value": 180.0,
  "attribute_5_value": 0.84
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  "copper_kg": 0.0,
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  "total_mass_kg": 6.7,
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  "attribute_4_value": 150.0,
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  "region": "Europe",
  "dominant material": "cu",
  "product": "EV Traction Motor Winding",
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"copper_kg": 9.5,
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"attribute_4_value": 190.0,
"attribute_5_value": 0.80
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  "product": "EV Traction Motor Winding",
  "copper_kg": 0.0,
  "aluminum_kg": 4.6,
  "total_mass_kg": 5.3,
  "non_material_cost_per_unit": 78.00,
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  "attribute_3_value": 0.55,
  "attribute_4_value": 155.0,
  "attribute_5_value": 0.65
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{
  "region": "India",
  "dominant material": "cu",
  "product": "EV Traction Motor Winding",
  "copper_kg": 5.0,
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  "total_mass_kg": 5.6,
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  "attribute_3_value": 0.50,
  "attribute_4_value": 145.0,
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  "region": "India",
  "dominant material": "al",
  "product": "EV Traction Motor Winding",
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  "aluminum_kg": 2.4,
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